Producer surplus in A = $\int_0^{P_{A0}} \left(p_{A0} - S_A(P) \right) dP$

Consumer surplus in A = $\int_0^{P_{A0}} \left(D_A(P) - p_{A0} \right) dP$

Consumer & producer surplus:

$$PS_A = (75-50)\cdot\left(100+\frac{1}{2}\cdot 75\right) = 3437.5 \text{ NOK}$$

$$CS_A = (150-75)\cdot\left(100+\frac{1}{2}\cdot 75\right) = 10312.5 \text{ NOK}$$

$$PS_B = (125-50)\cdot\left(100+\frac{1}{2}\cdot 150\right) = 13125 \text{ NOK}$$

$$CS_B = (150-125)\cdot\left(200+\frac{1}{2}\cdot 50\right) = 5625 \text{ NOK}$$

Total surplus: 32500 NOK

No transmission constraints:

$$PS = (100-50)\cdot\left(200+\frac{1}{2}\cdot 250\right) = 16250 \text{ NOK}$$

$$CS = (150-100)\cdot\left(300+\frac{1}{2}150\right) = 18750 \text{ NOK}$$

Total surplus: 35000 NOK

With 50 MW transfer capacity:

Price:

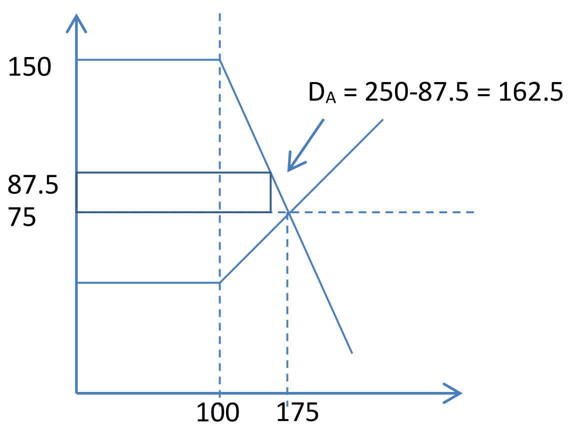
$$Demand: \quad p_A = 150 \qquad \text{for } D_A^{'} \leq 150$$
$$p_A = 300 - D_A^{'} \qquad \text{for } D_A^{'} \geq 150$$

With $D_A = S_a$ we get $300 - D_A^{'} = \frac{50}{3} + \frac{1}{3} D_A^{'}$

Resulting in $D_A^{'}$=212.5 and p_A = 87.5 (before: 75). D_A = 212.5 – 50 = 162.5

Correspondingly, p_B = 112.5

To calculate PS_A and CS_A you need to find demand in A for the price of 87.5:



$$PS_A = PS_{A0} + \left(87.5 - 75\right) \cdot (162.5 + \tfrac{1}{2} \cdot 12.5) + \tfrac{1}{2} \cdot 12.5 \cdot 50 =$$
$$3437.5 + 2109.375 + 312.5 = 5859.375 \text{ NOK}$$
$$CS_A = \left(150 - 87.5\right) \cdot \left(100 + \tfrac{1}{2} \cdot 62.5\right) = 8203.125 \text{ NOK}$$
or
$$CS_A = CS_{A0} - \Delta PS_A^{'} = 10312.5 - 2109.375 = 8203.125$$
where $\Delta PS_A^{'}$ is the change in PS_A excluding the export effect

Correspondingly:

Production in Area B with p_B = 112.5 is equal to: $S_B = 2 \cdot p_B = 225$

$$PS_B = \left(112.5 - 50\right) \cdot \left(100 + \tfrac{1}{2} \cdot 125\right) = 10156.25 \text{ NOK}$$
$$CS_B = CS_{B0} + \left(125 - 112.50\right) \cdot \left(225 + \tfrac{1}{2} \cdot 25\right) + \tfrac{1}{2} \cdot 12.5 \cdot 50 =$$
$$5625 + 2968.75 + 312.5 = 8906.25 \text{ NOK}$$
$$\text{Total surplus: } \mathrm{PS}_{A0} + \mathrm{CS}_{A0} + \mathrm{PS}_{B0} + \mathrm{CS}_{B0} + 2 \cdot 312.5 = 33125 \text{ NOK}$$

BUT THIS IS NOT ALL!

Congestion rent: interconnection creates additional value. Can go to SO, MO or others.

Congestion rent equal transfer multiplied with Δp: 50 · (112.5 – 87.5) = 1250 NOK

Figure 7.11 (welfare effects)

$$SurplusA = (S_A - D_A) = -300 + 4p_A$$
$$DeficitB = (D_B - S_B) = 500 - 4p_B$$

Can verify that these are equal for $p_A = p_B = 100$

Can calculate congestion rent for capacity T:

$$S_A - D_A = T \qquad -300 + 4p_A = T \qquad p_A = \frac{T+300}{4}$$

$$D_B - S_B = T \qquad 500 - 4p_B = T \qquad p_B = \frac{500-T}{4}$$

$$Congestion\,\mathrm{Re}nt: \qquad \left(\frac{500-T}{4} - \frac{300+T}{4}\right)\cdot T = 50T - 0.5T^2$$

Congestion rent is maximized by differentiation, find T=50

Generally maximized for T ≈ 0.5 times market demand

For the example, congestion rent = 50 · 25 = 1250, and welfare loss (of having insufficient interconnection capacity) equals 50 · 25 · 0.5 = 625.